



QpiAI™ - Joint AI Training and Certification Program

Feature Engineering, Hyperparameter Tuning

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Recap:

**Basic Probability &
Statistics**

**Regression and
Classification**

**Linear & Logistic
Regression**

**K-Nearest Neighbor
Algorithm**

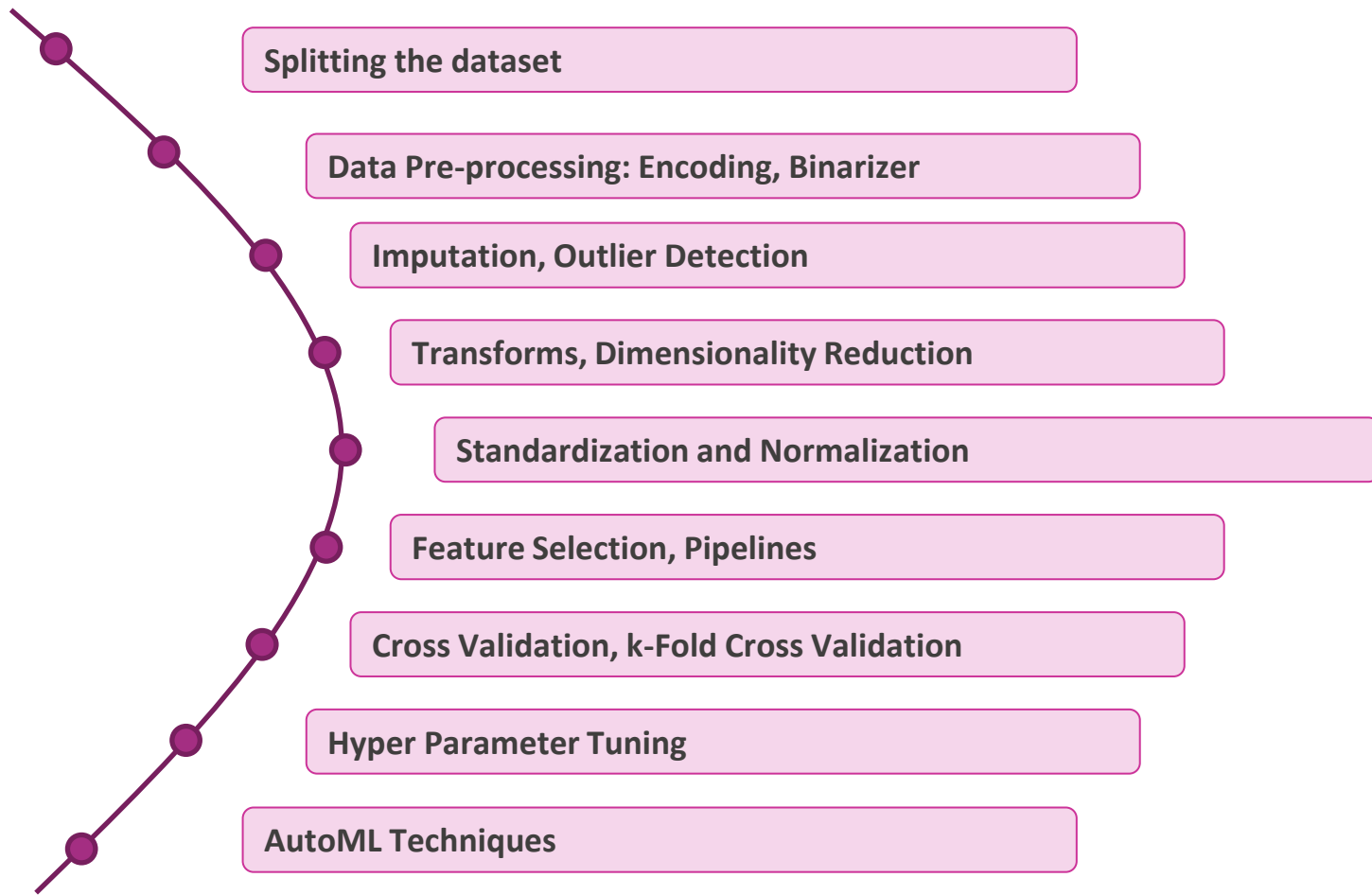
**Support Vector
Machines, Classification
Performance Metrics**

**Decision Tree,
Random Forest &
Ensemble Methods**

**Principal Component
Analysis and k- Means**

Time Series Analysis

Overview



KNN Imputation

This imputer utilizes the k-Nearest Neighbors method to replace the missing values in the datasets with the mean value from the parameter 'n_neighbors' nearest neighbours found in the training set. By default, it uses a Euclidean distance metric to impute the missing values.

Mean Imputation

Mean imputation is a method in which the missing value on a certain variable is replaced by the **mean** of the available cases.

Median Imputation

Median imputation consists of replacing all occurrences of missing values (NA) within a variable by its **median**.

Mode Imputation

Mode imputation replaces missing values of a categorical variable by the **mode or most common** of non-missing values of that variable.

Outlier Detection Through Interquartile

Helps us eliminate the outliers which are 1.5 times lower from the 25th percentile or 1.5 times above the 75th percentile. Works only with continuous values.

Outlier Detection Through Z-score

Helps eliminate the outliers which are 'n' standard deviations away from the mean. A column with continuous values should be given as inputs. Before applying this feature all columns should have values filled and there should be no infinite value.

Outlier Detection Through Isolation Forest

Helps eliminate multiple anomalies and outliers present in the dataset. It uses an algorithm similar to Random Forest. Multiple columns with continuous values can be given as inputs. Before applying this feature all columns should have values filled and there should be no infinite value. There are three types of Isolation Forest algorithms we have applied Isolation Forest Univariate, Isolation Forest Bi-variate, Isolation Forest Tri-variate.



$\sin \theta = \frac{OPP}{HYP}$
 $\cos \theta = \frac{ADJ}{HYP}$
 $\tan \theta = \frac{OPP}{ADJ}$
 $\csc \theta = \frac{HYP}{OPP}$
 $\sec \theta = \frac{HYP}{ADJ}$
 $\cot \theta = \frac{ADJ}{OPP}$

$1 = \sin^2 \theta + \cos^2 \theta$
 $1 = \sec^2 \theta - \tan^2 \theta$
 $1 = \csc^2 \theta - \cot^2 \theta$

$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}$
 $\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$
 $\tan \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$

$\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$
 $\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$
 $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$

$\sin \theta = \pm \frac{F_y}{F}$
 $\cos \theta = \pm \frac{F_x}{F}$
 $\tan \theta = \pm \frac{F_y}{F_x}$

$\vec{F} = F \cos \theta \hat{i} + F \sin \theta \hat{j}$
 $F_x = F \cos \theta$
 $F_y = F \sin \theta$

$\mu_0 = 4\pi \times 10^{-7} \frac{T \cdot m}{A}$
 $\epsilon_0 = 8.85 \times 10^{-12} \frac{C^2}{N \cdot m^2}$

$\oint \vec{E} \cdot d\vec{l} = -\frac{d\phi_E}{dt}$
 $\oint \vec{B} \cdot d\vec{l} = \mu_0 I + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$

$\oint \vec{E} \cdot d\vec{l} = (E + dE)h_f - Eh_f = h_f dE$
 $\phi_E = BA = Bh_f dx$
 $\frac{d\phi_E}{dt} = h_f dx \frac{dB}{dt}$
 $\oint \vec{E} \cdot d\vec{l} = -h_f dx \frac{dB}{dt}$
 $h_f dE = -h_f dx \frac{dB}{dt}$
 $\frac{dE}{dx} = -\frac{dB}{dt}$

$\frac{\partial^2 E}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2}$
 $E = E_{max} \sin(\omega t - kx)$

$\sin \theta = \frac{OPP}{HYP}$
 $\cos \theta = \frac{ADJ}{HYP}$
 $\tan \theta = \frac{OPP}{ADJ}$

$I_E = I_B + I_C$ and $\frac{I_C}{I_E} = \alpha$
 $I_B = I_E - I_C$
 $I_B = I_E (1 - \alpha)$
 $\beta = \frac{I_C}{I_B} = \frac{\alpha}{1 - \alpha}$

$V_{CC} = 10V$
 $R_1 = 1k\Omega$
 $R_2 = 1.5k\Omega$
 $R_3 = 100\Omega$
 $R_4 = 100\Omega$
 $R_5 = 2k\Omega$
 $C_1 = 10\mu F$
 $C_2 = 10\mu F$
 $C_3 = 490\mu F$
 $Q_1 = 2N3904$

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
 $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$

$M = \frac{f}{s - f} = \frac{s' - f}{f}$
 $\frac{1}{f} = \frac{1}{s} + \frac{1}{s'}$
 $m = \frac{i}{o} = \frac{s'}{s}$
 $f = \frac{R}{2}$

$6CO_2 + 12H_2O \xrightarrow{\text{light, Chlorophyll}} C_6H_{12}O_6 + 6O_2$
 $3GP \rightarrow RuBP \rightarrow Serine \rightarrow Glycine \rightarrow 2PGlycolate + 3PGlycolate$

$n \text{ HOOC(CH}_2)_4\text{COOH} + n \text{ H}_2\text{N(CH}_2)_6\text{NH}_2 \rightarrow \text{Nylon 6-6} + 2n \text{ H}_2\text{O}$

$\text{CH}_3\text{CH}_2\text{COOH}$ (Propanoic acid)
 $\text{CH}_3\text{CH}_2\text{OH}$ (Ethanol)

$\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3$ (Ethyl propanoate)

Thank you

Q&A